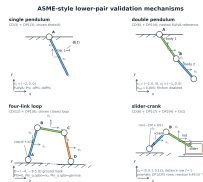


Verifier sandboxes, proof gates, and numerical test ladders for Lie-group multibody DAE methods

06/09/2026



Why this matters in our lab

Many people are attacking the same hard object:

**defensible multibody
time integration**

Lie groups
SO(3), SE(3), charts

**DAE
mechanisms**
joints, drivers,
reactions

friction/contact
smooth vs nonsmooth
tests

sparse solvers
AD, coloring, Newton

papers
benchmarks, claims,
refs

Agent purpose: keep method identity, examples, proof status, and paper claims synchronized across this shared evidence space.

Reference landscape drives the verifier design

stream	references	agent obligation
index-3 DAE	Hairer-Wanner; Gear-Leimkuhler-Gupta	check Φ , $\Phi_q v$, $\Phi_q a$, and multipliers
Lie-group time integration	Munthe-Kaas; Bruls-Cardona-Arnold; Wieloch-Arnold	preserve rotation geometry and method identity
time finite elements	Borri-Atluri; Hughes-Hulbert; Chaturvedi-Sandu-Sandu	separate source residual reproduction from local method claims
SBEL baselines	Taves-Kissel-Negrut; Kissel-Negrut-Taves; Fang et al.	make external rows comparable under explicit policy
friction/contact	Brown-McPhee; Stewart-Trinkle; Anitescu-Potra	separate smooth order from nonsmooth diagnostics

Why v001 had to become v048

v001-005	SO(3) order
v006-013	DAE closure
v014-022	friction smoothness boundary
v023-029	FullVA from endpoint consistency failures
v030-045	sparse scaling and colored AD
v046-048	ASME examples, proof ledger, external same-test policy

Evolution rule.

Each version exists because the previous verifier found a missing gate. The agent is a blocker-driven research controller, not a generic code loop.

All-method common-reference result matrix

Each cell reports observed velocity order p and finest-step velocity error e .

		Single	Double	Four-link	Slider-crank
Gauss6/FullVA	internal	$p=5.80$ $e=1.2e-12$	$p=6.01$ $e=2.3e-13$	$p=6.08$ $e=1.1e-11$	$p=7.34$ $e=9.2e-12$
HI22 rA	diagnostic	$p=1.00^*$ $e=7.7e-02$	$p=1.05^*$ $e=4.9e-05$	$p=1.02^*$ $e=2.0e-01$	$p=0.94^*$ $e=3.3e-02$
HI22 half	diagnostic	$p=1.84^*$ $e=7.7e-02$	$p=-0.80^*$ $e=3.0e+00$	$p=1.59^*$ $e=2.0e-01$	$p=2.46^*$ $e=3.3e-02$
RA21 rA	diagnostic	$p=0.62^*$ $e=1.3e+01$	$p=1.05^*$ $e=4.9e-05$	$p=1.02^*$ $e=2.0e-01$	$p=0.94^*$ $e=3.3e-02$
RA21 reps	diagnostic	$p=0.62^*$ $e=1.3e+01$	$p=1.05^*$ $e=4.9e-05$	$p=2.89^*$ $e=1.5e-02$	$p=0.94^*$ $e=3.3e-02$
RA21 rp	diagnostic	$p=0.52^*$ $e=1.5e+01$	$p=0.36^*$ $e=5.2e-05$	$p=2.89^*$ $e=1.5e-02$	$p=0.94^*$ $e=3.3e-02$
TFE Newmark	diagnostic	$p=0.90^*$ $e=1.5e+01$	$p=1.79^*$ $e=4.2e-04$	$p=1.87^*$ $e=1.7e-02$	$p=2.03^*$ $e=1.5e-03$
TFE m1	diagnostic	$p=1.02^*$ $e=1.5e+01$	$p=2.00^*$ $e=2.7e-04$	$p=3.87^*$ $e=5.3e-04$	$p=2.48^*$ $e=8.3e-04$
TFE m2	diagnostic	$p=0.05^*$ $e=1.1e+02$	$p=1.87^*$ $e=8.7e-05$	$p=1.02^*$ $e=6.7e-04$	$p=2.29^*$ $e=9.9e-03$
TFE trap.	diagnostic	$p=1.02^*$ $e=1.5e+01$	$p=2.00^*$ $e=2.7e-04$	$p=3.89^*$ $e=5.3e-04$	$p=2.33^*$ $e=7.3e-04$
VP24 rA proxy	diagnostic	$p=0.00^*$ $e=3.2e-11$	$p=1.00^*$ $e=2.3e-03$	$p=-0.00^*$ $e=1.1e+05$	$p=-0.00^*$ $e=1.3e-07$

■ $p \geq 5.5$ ■ $3 \leq p < 3.5$ ■ $1.5 \leq p < 3$ ■ $0.5 \leq p < 1.5$ ■ $p < 0.5$

* marks a row with diagnostic/source-policy caveats. Nonlocal rows are common-reference diagnostics only; they are not source-paper superiority claims.
Source: PAPER_NUMERICAL_RESULT_MATRIX.json; $h = 0.1, 0.05, 0.025$; reference $h = 0.0125$; 44/44 method-example cells.

Evidence matrix used to keep version status and claim
status aligned.

The agent we need

MethodSpec
what one-step map is
claimed

ProblemSpec
mechanisms and parameters

EvidenceSpec
CSV, JSON, plots, validators

ClaimSpec
accepted, diagnostic, open,
forbidden

generate

>

sandbox

>

prove

>

test

>

write

Human role: choose the research direction. Agent role: enforce gates before claims move.

Method object: the agent cannot verify a moving target

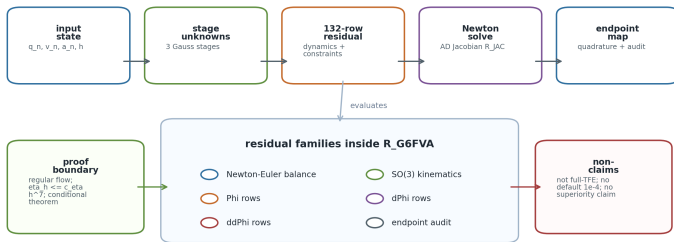
For v047

- 3 Gauss stages
- 132 FullVA rows
- position, velocity, acceleration constraints
- endpoint closure audited separately
- solver tolerance is a theorem gate

Non-claim: full source-paper TFE replacement unless its own residual gate closes.

Accepted Gauss6/FullVA one-step architecture

The figure separates the mathematical one-step map from open source-paper TFE replacement and external-superiority claims.



Accepted claim: a conditional sixth-order Gauss6/FullVA path for lower-pair mechanisms. Open gates remain explicit rather than hidden in the method diagram.

The method architecture is also the verifier target.

Verifier sandbox: constrain what the agent can run

level	sandbox	allowed question
L0	read-only validators	Do current paper, package, proof, and claim artifacts agree?
L1	target audit	Can one bounded hypothesis falsify this local gap?
L2	full generator	After method-code changes, do version-wide artifacts regenerate consistently?
L3	external source-policy	Under public same-test policy and opt-in, do external rows close?

Promotion rule: outputs move upward only when the lower sandbox cannot answer the claim question.

Proof sandbox: only gated facts enter the theorem

ID	obligation
P1	smooth chart and FullVA lift
P2	regular stage Jacobian and compact branch
P3	implemented residual defect $O(h^7)$
P4	endpoint closure perturbation $O(h^7)$
P5	Newton-Euler dynamic rows discharged
P6	solver residual $\eta_h \leq ch^7$
P7	residual-to-error theorem for surrogate rows

Accepted lane

direct residual bridge; AD row identity; Newton-Euler substitution; conditional sixth-order claim

Diagnostic lane

fixed tolerance logs; primitive Taylor fragments; external source-policy rows; residual-only rows

The ledger tells the agent what not to say.

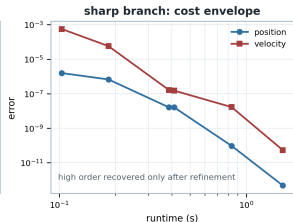
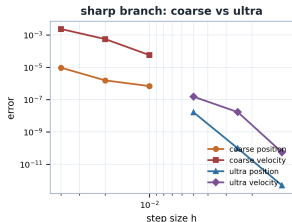
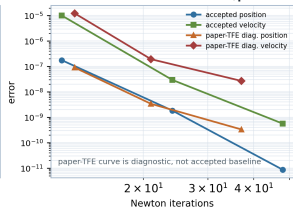
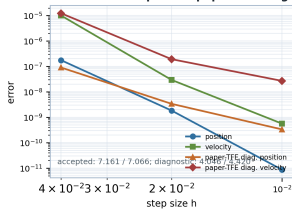
Numerical ladder: examples expose different failure modes

6	external same-test
5	ASME mechanisms
4	friction regimes
3	constrained DAE
2	smooth mechanics
1	SO(3) kinematics

Acceptance evidence: at least three h values, pinned reference policy, CSV/JSON/plot trail, and claim role.

Case evidence: smooth order verifier

Convergence and work/precision evidence for the accepted path
smooth branch: accepted vs paper-TFE diagnostic smooth branch: Newton work/precision



Accepted local method claim

- Gauss6 / FullVA
- conditional sixth-order path
- observed slopes about 7.16 / 7.07
- not an external superiority claim

Boundary: preserve this wording exactly.

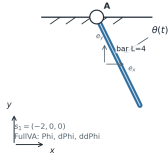
v047

Mechanism verifier: ASME lower-pair examples

ASME-style lower-pair validation mechanisms

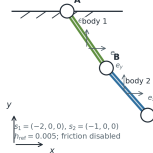
single pendulum

CD(3) + DP1(3); driven theta(t)



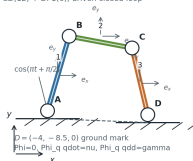
double pendulum

CD(6) + DP1(4); nested FullIVA reference



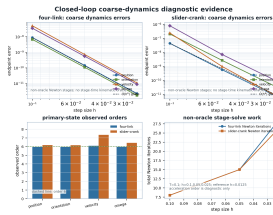
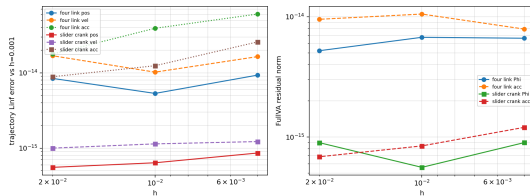
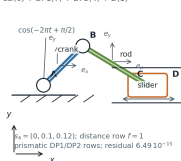
four-link loop

CD(12) + DP1(6); driven closed loop



slider-crank

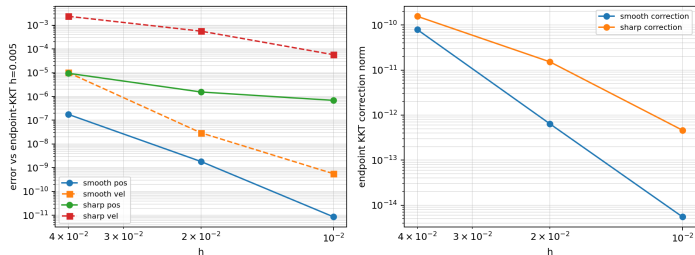
CD(6) + DP1(7) + DP2(4) + D(1)



Lower-pair graph bridge.

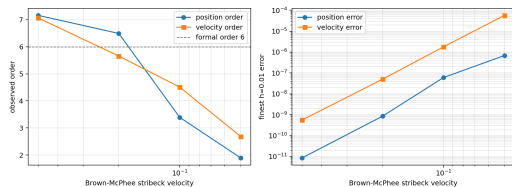
coverage and dynamic-order staging.

Endpoint closure is its own sandbox

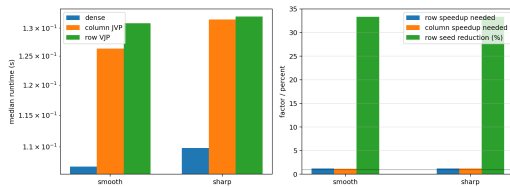


Why it matters. A projection can make constraints look clean while changing the one-step map. The agent cannot hide that inside an order claim.

Caveats are first-class evidence



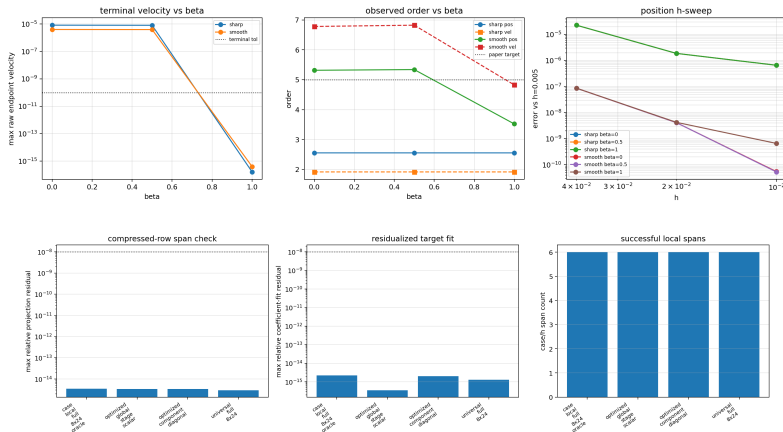
Friction smoothness boundary.



Sparse correctness versus sparse speed.

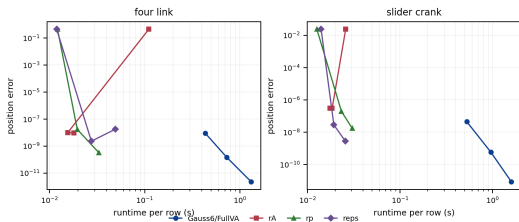
Agent behavior. Do not delete caveats. Promote smooth-order evidence, and demote sharp-friction and runtime caveats to their correct claim lanes.

Search strategy: negative results guide the next verifier

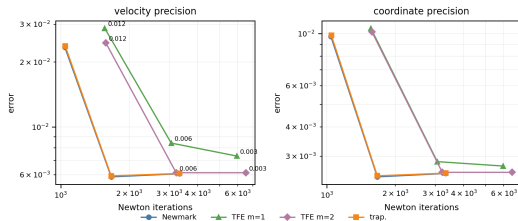


Terminal closure, source freedom, order preservation, span, rank, and residual are recorded separately.

External comparison remains bounded



Common-reference work/precision.



Literal TFE algorithm diagnostics.

Boundary. v048 provides same-test scaffolds, but source-policy external superiority remains false until strict public campaigns close.

Claim state machine

Claim boundary and remaining limitations

accepted method claim

accepted

Gauss6/FullVA order

conditional sixth-order path; observed smooth slopes 7.161/7.066 are finite-window support, not a seventh-order theorem

accepted

mechanism coverage

four ASME-style mechanisms are covered; dynamic order is accepted only for single and double pendula

bounded

source-paper comparison

local $m=3$ TFE formula target has expected order five; complete source-paper residual replacement is outside the claim

open

external superiority

not claimed until the same-test public-code campaign and work metrics are complete

proof and solver limitations

conditional

conditional theorem

requires regular reduced flow, smooth lift, isolated Newton solve, and stage residual defect $O(h^7)$

direct route

implementation oracle

runtime layout, AD-expanded oracle, and direct dynamic row closure pass; primitive symbolic lane remains diagnostic

open

nonlinear tolerance

proof-level policy is $\epsilon_{th} = O(h^7)$; fixed absolute tolerances are finite-run evidence

open

residual-to-error route

closed-loop residual rows are not promoted to order claims without the missing theorem

four-example evidence split



single

dynamic order accepted



double

dynamic order accepted



four-link

coarse dynamics / common-ref diagnostics; residual-to-error theorem open



slider-crank

coarse dynamics / common-ref diagnostics; residual-to-error theorem open

same-test benchmark limitations

open

original TFE pendulum

source-paper pendulum rows remain to be reproduced on the same policies

partial

2021 rA/rp/reps suite

public baselines and coarse pilots exist; full dynamic work/precision comparison remains open

partial

2022 half-implicit suite

bounded pilot is present; full T=8 policy and method rows remain open

open

velocity partitioning

code path unresolved in the local public-code mirror and metadata search

hypothesis

diagnostic

accepted

open

forbidden

A claim can move backward when a validator exposes stale prose or unfair policy.

How to build this agent for our integrator work

1

encode MethodSpec

2

pin ProblemSpec

3

choose sandbox level

4

run numerical ladder

5

update proof ledger

6

gate paper claims

Good autonomy here means fewer unverified claims, not fewer human research decisions.

References used by the agent design

DAE / Lie groups	Hairer & Wanner 1996; Gear, Leimkuhler & Gupta 1985; Munthe-Kaas 1998/1999; Bruls, Cardona & Arnold 2012; Wieloch & Arnold 2021.
TFE / higher order	Chaturvedi, Sandu & Sandu 2026, doi:10.1007/s11044-026-10153-w; Kim & Reddy 2017; Borri & Atluri 1988; Hughes & Hulbert 1988; Bauchau 2024.
SBEL baselines	Taves, Kissel & Negrut 2020; Kissel, Negrut & Taves 2021/2022; Fang et al. 2023; Kissel, Bakke & Negrut 2024.
Friction/contact	Brown & McPhee 2016; Stewart & Trinkle 1996; Anitescu & Potra 1997; Tasora & Anitescu 2011.
Local artifacts	main_cmame.tex bibliography; REFERENCE_METADATA_AUDIT.md/json; CURRENT_PIPELINE_CONTRACT.md; ORDER_PROOF_LEDGER.md; summary_v047.json; v048 same-test scaffolds.

Reference metadata audit status: 28 references; 16/16 DOI rows verified; 12/12 local non-DOI rows verified.